

CONTACT

Lehi, UT, United States
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[LinkedIn](#)
[Portfolio](#)

EDUCATION

Brigham Young University
Computer Science, B.S.

Minors: Korean, Math
2026 - 2029 | GPA: 3.95

• Brigham Young Scholarship recipient.

Skyridge High School
High School Diploma
2020 - 2023 | GPA: 4.0

• Falcon Scholar Award.
• 45 College Credit Hours.
• Highly involved in symphonic and marching band over 4 years.

SKILLS

Languages & Frameworks
Python, C++, HTML/CSS/JS, React, FastAPI, SQL

Tools & Platforms
Git/GitHub, Docker, PostgreSQL, Vercel, Render, Wix, Canva

Concepts
OCR, text parsing, REST APIs, authentication, full-stack development, AI integration, CI/CD pipelines

Languages
Korean (conversational proficiency)

TONY ADAIR

Computer Science Student | Full-Stack & Data-Focused Developer

TECHNICAL PROJECTS

RecipeLens

Personal Project | [github](#) | [live demo](#)

- Engineered and deployed a full-stack web application that converts recipe images into structured, multilingual recipe data using the Google Gemini API and Tesseract OCR.
- Built a Python/FastAPI backend supporting authentication, RESTful APIs, OCR extraction, and AI-assisted parsing workflows.
- Developed a React frontend featuring responsive UI components, drag-and-drop recipe editing, and persistent user collections.
- Implemented JWT authentication and migrated storage from SQLite to PostgreSQL using Supabase while deploying production infrastructure through Vercel and Render.
- Built a hybrid ingredient parsing system combining client-side regex extraction with Gemini API fallback to convert free-text recipe ingredients into structured data for scaling, search, and persistent storage.

Datalog Interpreter | C++

- Developed a recursive Datalog interpreter capable of evaluating relational queries across complex datasets.
- Implemented a custom lexer, parser, and relational algebra engine to process rules, facts, and query dependencies.
- Designed recursive query evaluation logic for handling multi-step data relationships and logical inference operations.

EXPERIENCE

Software Engineer Intern | Nightingale Heart Project

May 2026 – Present

- Resolved deployment and configuration issues in a Dockerized healthcare chatbot platform by repairing environment variable mismatches; successfully migrated the local PostgreSQL Tracker Store to Supabase cloud infrastructure to enable global user data aggregation.
- Collaborated in a team Git workflow using feature branches, pull requests, merge reviews, and repository synchronization practices.
- Refactored NLU training schemas and containerized pipelines using Rasa and Docker to disambiguate user dialogue pathways, reducing intent fallback rates across 100+ distinct conversational edge cases.
- Facilitated the onboarding of a new team member working in the same role.

Full-Time Missionary in South Korea | The Church of Jesus Christ of Latter-Day Saints

Sep 2023 – Sep 2025

- Conducted full-time community outreach and teaching in South Korea, engaging daily with individuals from diverse cultural and linguistic backgrounds.
- Taught ESL in one-on-one and group settings using structured curricula.
- Mentored and trained 5-12 volunteers in leadership roles, improving team coordination and performance.
- Communicated fluently in Korean during teaching, mentoring, and public-facing interactions.
- Built relationships and resolved interpersonal conflicts across cultural and linguistic barriers.
- Developed and produced digital media content for a mission social media team, leading video projects from concept to publication and serving as a trusted resource for content creators and team leadership on production strategy and best practices.

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ADDITIONAL EXPERIENCE

Undergraduate Research Assistant | Human-Centered Machine Intelligence Lab at BYU

Aug 2026 – Present

AI Systems & Automation Intern | Strong Force Analytics

May 2026 – Aug 2026

- Built an agentic Python pipeline that transforms natural-language prompts into playable HTML5 Canvas games using Gemini, Groq, and Ollama models.
- Developed an automated QA and repair system combining Playwright, static analysis, runtime telemetry, and LLM evaluation to detect and iteratively fix generated game defects.
- Designed orchestration logic that classified failures and selected between targeted code repairs, regeneration, or design revisions.
- Implemented a PostgreSQL-backed telemetry system for storing execution history, enabling data-driven improvements to generation and repair strategies.
- Developed a retrieval-based semantic search chatbot that indexes website content with vector embeddings using RAG techniques.

Teaching Assistant | Brigham Young University — CS 236: Discrete Structures

June 2026 – Aug 2026

- Graded homework assignments and provided constructive feedback to help students develop their understanding of discrete structures and computer science concepts.
- Assisted students with questions and troubleshooting related to programming projects, labs, and coursework, guiding them toward solutions and stronger problem-solving skills.

Operations Assistant | Brigham Young University

Oct 2025 – Aug 2026

- Executed large-scale relocation of campus equipment, supplies, and furnishings across multiple facilities, including inventorying, organizing, cleaning, and transporting hundreds of items during a major building closure.
- Collaborated with operations staff to manage evolving priorities, maintain organized inventories, and efficiently coordinate daily relocation and facility operations with minimal supervision.

Business Systems Assistant | Thrive 55+ Nursing Advantage

Oct 2025 – June 2026

- Established foundational digital business systems for a small healthcare-focused company, including website development, CRM setup, and client-facing operational infrastructure.
- Designed digital workflows and branding materials using Wix and Canva to support client outreach, online presence, and customer acquisition efforts.